



1717 Wakonade Drive  
Welch, MN 55089

October 22 , 2024

L-PI-24-047  
10 CFR 50.73

ATTN: Document Control Desk  
U.S. Nuclear Regulatory Commission  
Washington, DC 20555-0001

Prairie Island Nuclear Generating Plant, Units 1 and 2  
Docket Nos. 50-282 and 50-306  
Renewed Facility Operating License Nos. DPR-42 and DPR-60

Prairie Island Nuclear Generating Plant Licensee Event Report 2024-001-01

Northern States Power Company, a Minnesota corporation, doing business as Xcel Energy (hereafter "NSPM"), hereby submits Licensee Event Report (LER) 50-282/2024-001-01 per 10 CFR 50.73(a)(2)(i)(B). This report is a planned supplement to LER 2024-001-00 submitted on September 16, 2024 (ADAMS Accession number ML24260A256).

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.

A handwritten signature in blue ink, appearing to read 'Bryan Currier', is written over a printed name.

Bryan Currier  
Plant Manager, Prairie Island Nuclear Generating Plant  
Northern States Power Company – Minnesota

Enclosure

cc: Administrator, Region III, USNRC  
Project Manager, Prairie Island, USNRC  
Resident Inspector, Prairie Island, USNRC  
State of Minnesota

**ENCLOSURE**

**PRAIRIE ISLAND NUCLEAR GENERATING PLANT  
LICENSEE EVENT REPORT 50-282/2024-001-01**

4 pages follow



## LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to [Infocollections.Resource@nrc.gov](mailto:Infocollections.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

|                                                                     |                                         |                          |                   |
|---------------------------------------------------------------------|-----------------------------------------|--------------------------|-------------------|
| 1. Facility Name<br>Prairie Island Nuclear Generating Plant, Unit 1 | <input checked="" type="checkbox"/> 050 | 2. Docket Number<br>-282 | 3. Page<br>1 OF 4 |
|                                                                     | <input type="checkbox"/> 052            |                          |                   |

|                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| 4. Title<br>Auxiliary Building Special Ventilation System Inoperable during movement of irradiated fuel assemblies. |
|---------------------------------------------------------------------------------------------------------------------|

| 5. Event Date |     |      | 6. LER Number |                   |              | 7. Report Date |     |      | 8. Other Facilities Involved |                                         |               |
|---------------|-----|------|---------------|-------------------|--------------|----------------|-----|------|------------------------------|-----------------------------------------|---------------|
| Month         | Day | Year | Year          | Sequential Number | Revision No. | Month          | Day | Year | Facility Name                | <input checked="" type="checkbox"/> 050 | Docket Number |
| 05            | 02  | 2024 | 2024          | - 001 -           | 01           | 10             | 22  | 2024 | Prairie Island Unit 2        |                                         | -306          |
|               |     |      |               |                   |              |                |     |      | Facility Name                | <input type="checkbox"/> 052            | Docket Number |

|                        |                        |
|------------------------|------------------------|
| 9. Operating Mode<br>1 | 10. Power Level<br>100 |
|------------------------|------------------------|

## 11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

|                                                                          |                                                    |                                                       |                                             |                                                    |                                     |
|--------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------|---------------------------------------------|----------------------------------------------------|-------------------------------------|
| <input checked="" type="checkbox"/> 10 CFR Part 20                       | <input type="checkbox"/> 20.2203(a)(2)(vi)         | <input checked="" type="checkbox"/> 10 CFR Part 50    | <input type="checkbox"/> 50.73(a)(2)(ii)(A) | <input type="checkbox"/> 50.73(a)(2)(viii)(A)      | <input type="checkbox"/> 73.1200(a) |
| <input type="checkbox"/> 20.2201(b)                                      | <input type="checkbox"/> 20.2203(a)(3)(i)          | <input type="checkbox"/> 50.36(c)(1)(i)(A)            | <input type="checkbox"/> 50.73(a)(2)(ii)(B) | <input type="checkbox"/> 50.73(a)(2)(viii)(B)      | <input type="checkbox"/> 73.1200(b) |
| <input type="checkbox"/> 20.2201(d)                                      | <input type="checkbox"/> 20.2203(a)(3)(ii)         | <input type="checkbox"/> 50.36(c)(1)(ii)(A)           | <input type="checkbox"/> 50.73(a)(2)(iii)   | <input type="checkbox"/> 50.73(a)(2)(ix)(A)        | <input type="checkbox"/> 73.1200(c) |
| <input type="checkbox"/> 20.2203(a)(1)                                   | <input type="checkbox"/> 20.2203(a)(4)             | <input type="checkbox"/> 50.36(c)(2)                  | <input type="checkbox"/> 50.73(a)(2)(iv)(A) | <input type="checkbox"/> 50.73(a)(2)(x)            | <input type="checkbox"/> 73.1200(d) |
| <input type="checkbox"/> 20.2203(a)(2)(i)                                | <input checked="" type="checkbox"/> 10 CFR Part 21 | <input type="checkbox"/> 50.46(a)(3)(ii)              | <input type="checkbox"/> 50.73(a)(2)(v)(A)  | <input checked="" type="checkbox"/> 10 CFR Part 73 | <input type="checkbox"/> 73.1200(e) |
| <input type="checkbox"/> 20.2203(a)(2)(ii)                               | <input type="checkbox"/> 21.2(c)                   | <input type="checkbox"/> 50.69(g)                     | <input type="checkbox"/> 50.73(a)(2)(v)(B)  | <input type="checkbox"/> 73.77(a)(1)               | <input type="checkbox"/> 73.1200(f) |
| <input type="checkbox"/> 20.2203(a)(2)(iii)                              |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(A)            | <input type="checkbox"/> 50.73(a)(2)(v)(C)  | <input type="checkbox"/> 73.77(a)(2)(i)            | <input type="checkbox"/> 73.1200(g) |
| <input type="checkbox"/> 20.2203(a)(2)(iv)                               |                                                    | <input checked="" type="checkbox"/> 50.73(a)(2)(i)(B) | <input type="checkbox"/> 50.73(a)(2)(v)(D)  | <input type="checkbox"/> 73.77(a)(2)(ii)           | <input type="checkbox"/> 73.1200(h) |
| <input type="checkbox"/> 20.2203(a)(2)(v)                                |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(C)            | <input type="checkbox"/> 50.73(a)(2)(vii)   |                                                    |                                     |
| <input type="checkbox"/> OTHER (Specify here, in abstract, or NRC 366A). |                                                    |                                                       |                                             |                                                    |                                     |

## 12. Licensee Contact for this LER

|                                   |                                                  |
|-----------------------------------|--------------------------------------------------|
| Licensee Contact<br>Nathan Fedora | Phone Number (Include area code)<br>612-342-8971 |
|-----------------------------------|--------------------------------------------------|

## 13. Complete One Line for each Component Failure Described in this Report

| Cause | System | Component | Manufacturer | Reportable to IRIS | Cause | System | Component | Manufacturer | Reportable to IRIS |
|-------|--------|-----------|--------------|--------------------|-------|--------|-----------|--------------|--------------------|
|       |        |           |              |                    |       |        |           |              |                    |

## 14. Supplemental Report Expected

|                                        |                                                                              |                              |       |     |      |
|----------------------------------------|------------------------------------------------------------------------------|------------------------------|-------|-----|------|
| <input checked="" type="checkbox"/> No | <input type="checkbox"/> Yes (If yes, complete 15. Expected Submission Date) | 15. Expected Submission Date | Month | Day | Year |
|                                        |                                                                              |                              |       |     |      |

## 16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On May 02, 2024, the Prairie Island Nuclear Generating Plant (PINGP) Units 1 and 2 were operating at 100 percent power while performing fuel handling operations. For a 33-minute period between 12:52 and 13:25, radiation monitors 1R-37 and 2R-30 were both out of service while movement of irradiated fuel assemblies was occurring. On July 11, 2024, while completing a review of the design and licensing basis of radiation monitors 1[2]R-37, 1[2]R-30, it was determined that these monitors are required for Auxiliary Building Special Ventilation System (ABSVS) operability during movement of irradiated fuel assemblies per Technical Specification (TS) 3.7.12. Changes to fuel handling procedures have been initiated. TS Bases for the ABSVS were further clarified and communicated to the operating crews. This will ensure compliance with TS going forward when irradiated fuel is being moved. This event is reportable under 10 CFR 50.73(a)(2)(i)(B) for an Operation or Condition Prohibited by Technical Specifications.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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|                                                                                |                                                       |                                     |                      |                                   |                        |
|--------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------|----------------------|-----------------------------------|------------------------|
| <b>1. FACILITY NAME</b><br><br>Prairie Island Nuclear Generating Plant, Unit 1 | <input checked="checked" type="checkbox"/> <b>050</b> | <b>2. DOCKET NUMBER</b><br><br>-282 | <b>3. LER NUMBER</b> |                                   |                        |
|                                                                                | <input type="checkbox"/> <b>052</b>                   |                                     | <b>YEAR</b><br>2024  | <b>SEQUENTIAL NUMBER</b><br>- 001 | <b>REV NO.</b><br>- 01 |

**NARRATIVE****Plant Operating Conditions Prior to the Event:**

Unit 1 – Mode 1, 100 percent Power.

Unit 2 – Mode 1, 100 percent Power.

**EVENT DESCRIPTION**

On May 02, 2024, the Prairie Island Nuclear Generating Plant (PINGP) was performing fuel handling operations in the Spent Fuel Pool area in support of a Cask loading campaign. For a 33-minute period between 12:52 and 13:25, Radiation Monitors [IL] 1R-37 (Unit 1, Train A) and 2R-30 (Unit 2, Train B) were both out of service while movement of irradiated fuel assemblies was occurring. This impacted the Auxiliary Building Special Ventilation System [VF] (ABSVS) and caused both Train A and Train B of the ABSVS to be inoperable during the movement of irradiated fuel assemblies. The implications of this condition were not known at the time.

Radiation monitors 1[2]R-37, 1[2]R-30 are vent gas monitors that monitor for potential radiological releases out of the Auxiliary Building Normal Ventilation (ABNV) exhaust stacks. One of the safety functions of these monitors is to send an auto-start signal to the ABSVS upon detecting high radiation. The Unit 1 ABNV exhaust stack is monitored by 1R-37 (Train A) and 1R-30 (Train B) while the Unit 2 ABNV stack is monitored by 2R-37 (Train A) and 2R-30 (Train B). The ABSVS is a single system that serves both Unit 1 and Unit 2. The system consists of two redundant trains: Train A and Train B. Actuation of either train of the ABSVS will isolate the ABNV exhaust stacks on both Units.

The high radiation signal from 1[2]R-37 and 1[2]R-30 is the credited ABSVS actuation signal during a Fuel Handling Accident (FHA) event. With 1R-37 out of service (OOS), the A Train ABSVS would not receive this auto-start signal from a release through the Unit 1 ABNV exhaust stack thus making Train A inoperable during the movement of irradiated fuel assemblies. With 2R-30 OOS, the B train ABSVS would not receive this auto-start signal from a release through the Unit 2 ABNV exhaust stack thus making Train B inoperable during the movement of irradiated fuel assemblies.

Technical Specification (TS) Limiting Condition for Operation (LCO) 3.7.12 Auxiliary Building Special Ventilation System (ABSVS) requires "Two ABSVS trains shall be OPERABLE" and is applicable in MODES 1, 2, 3 and 4 and "During movement of irradiated fuel assemblies". The radiation monitors only impact the operability of an ABSVS train during movement of irradiated fuel assemblies.

There are no conditions in TS 3.7.12 that apply to two Trains of ABSVS being inoperable during movement of irradiated fuel assemblies due to radiation monitors out of service. With no applicable action provided in Tech Spec 3.7.12, LCO 3.0.3 applied since LCO 3.0.3 establishes the actions that must be implemented when an LCO is not met and an associated action is not provided. LCO 3.0.3 states that the unit shall be placed in a MODE or other specified condition in which the LCO is not applicable and delineates the time limits for placing the unit in a safe MODE. However, since the high radiation signal is only required for LCO 3.7.12 during movement of irradiated fuel, a unit shutdown would not provide appropriate remedial measures. The appropriate action in response to two trains of ABSVS being inoperable during movement of irradiated fuel assemblies due to radiation monitors out of service would be to immediately suspend the movement of irradiated fuel assemblies, but that did not happen.

This event is reportable under 10 CFR 50.73(a)(2)(i)(B) for an Operation or Condition Prohibited by Technical Specifications.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

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|                                                                                |                                                |                                     |                      |                                   |                        |
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|                                                                                | <input type="checkbox"/> <b>052</b>            |                                     | <b>YEAR</b><br>2024  | <b>SEQUENTIAL NUMBER</b><br>- 001 | <b>REV NO.</b><br>- 01 |

**NARRATIVE****DATES AND APPROXIMATE TIMES OF MAJOR OCCURRENCES**

October 19, 2023, 1R-37 Auxiliary Building Ventilation Gas Radiation Monitor was OOS due to a failed sample pump. With 1R-37 OOS, the A Train ABSVS would not receive a start signal on high radiation out of the Unit 1 ABNV exhaust stack. 1R-37 remained OOS for an extended period.

February 13, 2024, a change to the Technical Specification Bases was completed to clarify that the high radiation initiation of ABSV is required to trip the ABNVS.

May 02, 2024 [Event Date] from 12:52 to 14:59, 2R-30 was OOS for calibration. During this time, B train ABSVS would not receive a start signal on high radiation out of the Unit 2 ABNV exhaust stack.

May 02, 2024 movement of irradiated fuel took place from 0820 to 1325.

July 11, 2024, while completing a review of the design and licensing basis of radiation monitors 1[2]R-37 and 1[2]R-30, it was identified that the issues with 1R-37 out of service may impact ABSVS operability per Technical Specification (TS) 3.7.12 while moving irradiated fuel assemblies.

July 26, 2024 [Discovery Date] a Past Operability Review (POR) was completed which reviewed the time periods when movement of irradiated fuel occurred at the same time as one or more of the associated radiation monitors were out of service to see if a past inoperability of the ABSVS occurred. The POR identified numerous occurrences when one or more of the radiation monitors were out of service during movement of irradiated fuel.

August 13, 2024, a concern was raised questioning the conclusion of the POR. It was identified that the POR incorrectly concluded operability of the ABSVS was maintained throughout these identified occurrences. A revision to the POR was then initiated.

August 20, 2024, the site completed a revision of the POR. In this revision, the site correctly identified the periods of time of inoperability of the ABSVS due to the associated radiation monitors being out of service. One of the instances identified was a period when two radiation monitors were out of service impacting both Train A and Train B of the ABSVS. The revised POR concluded that both Train A and Train B ABSVS were inoperable during movement of irradiated fuel assemblies on May 02, 2024 from 12:52 to 13:25 for a period of 33 minutes.

It was not recognized that LCO 3.0.3 was applicable for the condition the plant was in, and the appropriate action in response to two Trains of ABSVS being inoperable during movement of irradiated fuel assemblies due to radiation monitors out of service would have been to immediately suspend the movement of irradiated fuel assemblies, but that did not happen.

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|-------------------------------------------------|-----------------------------------------|------------------|---------------|-------------------|---------|
| Prairie Island Nuclear Generating Plant, Unit 1 | <input checked="" type="checkbox"/> 050 | -282             | YEAR          | SEQUENTIAL NUMBER | REV NO. |
|                                                 | <input type="checkbox"/> 052            |                  | 2024          | - 001             | - 01    |

**NARRATIVE****ASSESSMENT OF SAFETY CONSEQUENCES**

During this event, there was always a redundant radiation monitor capable of detecting a potential radiological release out of either unit's exhaust stack which would have resulted in a Train A or Train B ABSVS actuation during a postulated FHA depending on which unit's exhaust stack the high radiation was first detected. Actuation of either the Train A or Train B ABSVS results in the elimination of both Unit 1 and Unit 2 ABNV exhaust stacks as potential radiological release sources. This performs the safety function credited by the ABSVS during an FHA event.

The health and safety of the public and site personnel were not impacted during this event.

**CAUSE OF THE EVENT**

TS Bases were changed to clarify operability requirements of the ABSV system without ensuring the license holders understood the changes to TS 3.7.12 Bases.

Operating procedures were not updated to include steps to evaluate TS 3.7.12 if radiation monitors were inoperable or out of service.

**CORRECTIVE ACTIONS**

TS Bases for the ABSV system were further clarified and communicated to the operating crews.

Changes to fuel handling procedures have been initiated. This will ensure compliance with TS going forward when irradiated fuel is being moved.

**PREVIOUS SIMILAR EVENTS**

There were no previous similar events in the past three years.

**ADDITIONAL INFORMATION**

All times are in Central Daylight Time.

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].